

I-CHUN CHEN

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🔗 <https://github.com/wazenmai>

🔗 <https://blog.wazenmai.com>

My research interests broadly lie in the fields of multimodal (image, text, and music) AI, model compression, and explainable AI. Alongside my background spanning full-stack iOS & Android mobile app development and C++/Python inference pipelines, I have a first-author paper accepted at **ICML 2025** exploring Mixture-of-Experts (MoE) LLM compression. Currently, I apply this expertise to develop **agent harnesses** and advance text-to-speech models.

EDUCATION

National Tsing Hua University

September 2022 - July 2024

Master's degree. Department of Computer Science.

- **Overall GPA:** 4.23/4.3

Technische Universität Dresden, Germany

March 2022 - August 2022

Exchange Student of Computer Science.

National Tsing Hua University

September 2018 - August 2022

Bachelor's degree. Department of Computer Science.

- **Overall GPA:** 3.93/4.3

PUBLICATIONS

Retraining-Free Merging of Sparse Mixture-of-Experts via Hierarchical Clustering

🔗 <https://arxiv.org/abs/2410.08589>

🔗 <https://github.com/wazenmai/HC-SMoE>

- Citation count: 19, Github star: 25.

- Accepted by Forty-Second International Conference on Machine Learning (ICML 2025) as first author.

- Reduced expert parameters in MoE models using a retraining-free, task-agnostic approach.

- Utilized hierarchical clustering to group similar experts, completing the merging process in under ten minutes.

- Evaluate Mixtral 8x7B and Qwen1.5-MoE on eight language benchmarks.

MidiBERT-Piano: Large-scale Pre-training for Symbolic Music Understanding

🔗 <https://arxiv.org/abs/2107.05223>

🔗 <https://github.com/wazenmai/MIDI-BERT>

- Citation count: 100+, Github star: 200+.

- Accepted by Journal of Creative Music Systems (JCMS), 2024.

- Applied BERT to general musical tasks, including melody completion, composer classification, and emotion classification using PyTorch.

- Evaluated performance differences between REMI and CP music representations.

WORK EXPERIENCE

Cyberlink

December 2024 - Now

Senior Software Engineer (AI-team)

- Keywords: *Agent Harnesses, Vision-Language Models (VLMs), Evaluation Systems, Text-to-Speech (TTS), Computer Vision.*

- Trained a human parsing model based on Segformer that achieves mIoU 77.03 on the validation set of DeepFashion.

- Implemented OpenVINO and ONNXRuntime C++ inference pipeline for human parsing.

- Evaluated state-of-the-art VLMs (DeepSeek, QwenVL-2, Ola, Ovis-2) and improved inference throughput by 1.6x using vLLM.

- Implemented auto quality evaluator by VLM(Gemma-3) in Python.

- Constructed an automated video storytelling pipeline using Python, FFmpeg, Gemini (VLM), and IndexTTS.

- Designed loss function for image harmonization on shadow generation.

- Optimized IndexTTS inference time (improving RTF from 0.4 to 0.15 for 120s audio) and enhanced adaptability for Traditional Chinese (zh-TW).

Cloud Communication Technology LTD.

February 2022 - August 2023

iOS and Android App Engineer

- Implemented LINE message API and login service in Community Web Flutter app.
- Implemented version detection, encrypted qrcode scanner, Google Cloud Storage Image upload and deeplink function in TABF Exam Camera Flutter app.
- Embedded a Google face detector into a Flutter camera app to enable real-time detection of face angles and positions for ID photos.
- Designed and implemented backend APIs and database CRUD operations for the ID Photo camera using TypeScript, Koa, and PostgreSQL.

Research Center for Information Technology Innovation, Academia Sinica

July 2020 - August 2020

Music & AI Lab Summer Intern

- Learned symbolic music format representation and encoding.
- Presented papers in NLP transformer family, including Transformer, BERT, RoBERTa, ELECTRA, GPT-2, and T5.
- Upgraded 'Pop Music Transformer' from single-track to multi-track music generation, enabling the model to generate original songs or synthesize continuations from human-provided melodies.

PROJECTS

2022 AI-CUP Explanation Information Tagging of Natural Language Understanding Competition

🔗 <https://github.com/wazenmai/DataMining-Team5>

- Analyzed conversational transcripts and speaker sentiment to identify key statements that support their main arguments.
- Framed the problem as a token classification task, fine-tuning BERT, DistilBERT, and RoBERTa models, as well as a question-answering task using BERT."
- Achieved 83.6% accuracy using a token classification RoBERTa model, ranking 30th on the private leaderboard.

NTHU OAuth Decaptcha

🔗 <https://chrome.google.com/webstore/detail/nthu-oauth-decaptcha/mflpajkffpiibelpmffonlenndbgogp>

🔗 <https://github.com/wazenmai/NTHU-OAuth-Decaptcha>

- Trained a CNN model to recognize the number on the captcha image, get 98.9% accuracy on testing set.
- Automated training data collection by scraping CAPTCHA images via HTTP requests.
- There are 356 users using our decaptcha extension.

Parallel Pitch Tracking Algorithm

🔗 <https://github.com/wazenmai/Parallel-Pitch-Tracking>

- Implemented an Auto-Correlation Function (ACF) using MPI and CUDA for pitch recognition in WAV files.
- About 100x speedup with the largest testing data by CUDA optimization.

TECHNICAL SKILLS

Programming:	C, C++, Python, Dart, Javascript, CUDA
Software & Tools:	ML-Related Framework: Pytorch, Huggingface-transformers
	Frontend Framework: Flutter, VueJS
	Cloud Service: GCP(Vision API, Cloud Storage), Firebase
	Database: PostgreSQL, MongoDB

EXTRACURRICULAR

Machine Learning Study Group

November 2019 - August 2021

- Participated in weekly discussions on ML concepts, implemented core algorithms, and reviewed academic papers.
- Including course of Machine Learning on Coursera by Andrew Ng, CS231n: Deep Learning of Computer Vision at Stanford University, Reinforcement Learning course by DeepMind David Silver, and CS294-112: Deep Reinforcement Learning at UC Berkely by Sergey Levine.

LeetCode Study Group - Convener

February 2021 - June 2021

- Organized and led a 16-student study group focused on topical LeetCode problem-solving.
- Got **the Excellent Award** of NTHU 2021 winter semester study group.

🔗 <https://justin0u0.notion.site/LeetCode-57d3fc3c39714bf7b37429a590a85824>